Supplement: Methods and Results for Direct Elicitation versus Ranked Nodes

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1 Construction and evaluation details

This section complements the main article's Method with the numerical specification and retrieval settings. It describes the original experimental pipeline, distinct from the alternative fits reported below.

1.1 Scrum reference reconstruction

The evaluated XML stores ranked-node functions, weights, variances, and expressions. We reconstructed its 249 conditional rows using equal-width intervals and five endpoint-inclusive points per parent state, preserving two piecewise expressions and the storage order of a volatility variable. The archived implementation has 42 nodes and three states, whereas the published framework describes 44 nodes and five states. The reconstructed tables are not an independently certified AgenaRisk export. Other references are stored CPTs; SESSI uses the original elicited table in Appendix A3 of its source publication.

1.2 Ranked-node equations and numerical fitting

Ordered states divide $[0, 1]$ into equal intervals. Negative influences reverse semantic rank before aggregation; predictions are mapped back to the stored child order. Anchors include all-low, all-high, all-middle (rank $\lfloor (c-1)/2 \rfloor$ for each parent's c states), each one-high/rest-low configuration, and each one-low/rest-high configuration, with duplicates removed. The anchor selection consults no reference probabilities.

For k parents, let G_r be the Cartesian product of five equally spaced, endpoint-inclusive points within each parent interval for row r. With standard-normal CDF Φ, child interval $[a_s, b_s]$, aggregation mean μ, and variance v, define

$$T_s(\mu, v) = \frac{\Phi((b_s - \mu)/\sqrt{v}) - \Phi((a_s - \mu)/\sqrt{v})}{\Phi((1-\mu)/\sqrt{v}) - \Phi(-\mu/\sqrt{v})}. \tag{1}$$

The predicted probability is $q_{rs} = 5^{-k} \sum_{x \in G_r} T_s(f(x; w), v)$. Each truncated distribution is normalized before averaging. The mean functions are

$$f_{\text{mean}}(x; w) = \frac{\sum_i w_i x_i}{\sum_i w_i}, \tag{2}$$

$$g_i(x; w) = \frac{w_i x_i + \sum_{j \neq i} x_j}{w_i + k - 1}, \tag{3}$$

$$f_{\min} = \min_i g_i, \qquad f_{\max} = \max_i g_i, \tag{4}$$

$$f_{\text{mix}}(x; a) = (1-a) \min_i x_i + a \max_i x_i. \tag{5}$$

For a single parent, the mean is its aligned value x_1. Weighted-mean log weight ratios relative to the last parent lie in $[-6, 6]$; weighted-minimum/maximum weights lie in $[1, 20]$; $a \in [0, 1]$; and $v \in [10^{-4}, 1]$. Variance and minimum/maximum weights use logarithmic parameterizations.

For the family selected by the LLM, the fitter minimized mean anchor total variation between fitted and elicited distributions. It used eight deterministic starts (seed 42), coordinate-search steps 2^{-2} through 2^{-11} on normalized parameter coordinates, at most 20 sweeps per step, and an improvement threshold of 10^{-12}. Ties retained the earlier

candidate. This seed controls fitting, not LLM sampling. The fitter received no reference CPT input. Poor but valid fits were retained; similar tables can arise from different parameters, so agreement does not establish parameter recovery.

1.3 Evidence retrieval and elicitation controls

Before collection, retrieval filtered evidence cards by model context and scored overlap between their variable tags and the target-plus-parent set; general-purpose tags contributed 0.25. It selected up to three positive-scoring cards, breaking ties by identifier. Packets contained 24–103 words (median 72), within a 220-word budget, and were fixed across paired conditions and replications. Cards excluded reference probabilities, priors, weights, variances, numerical source results, and links identifying reference artifacts. Both evidence arms received node definitions. Input-token budgets were not equalized.

All judgments conditioned on all listed parents and averaged over unspecified circumstances, without fixing omitted circumstances to middle states or assuming independence. Structured prompts framed probabilities as expected counts among 100 hypothetical comparable cases, allowed fractional counts, and requested consistency and normalization checks. These hypothetical cases supplied no observations. A brief evidence note was requested, without collecting private reasoning traces. Influence assumptions were shared between methods, but enforced mathematically only by RNM: nine followed from definitions, 43 had qualitative support, and 40 were uncertain hypotheses. Single-parent means used the direction-aligned parent value without selecting a combination rule.

1.4 Additional outcomes and inference

Secondary calculations retain Jensen–Shannon divergence in bits, mean absolute ordinal-CDF differences, and smoothed row-weighted CPT KL divergence. KL adds 10^{-8} to each state and renormalizes; its row weighting differs from the primary macro-average and it is not joint-distribution KL. TV remains the primary endpoint. Complete requested row sets and actual request order are available in the replication materials.

Replication blocks, rather than correlated targets, were paired for inference. The paired-t test concerns the mean difference, whereas the exact sign test concerns direction/median under independence. Fresh sessions and shuffled order do not establish normality, independence, or stationarity. Separate Holm families provide exploratory family-wise adjustment, not study-wide confirmatory control. Intervals are pointwise. No smallest effect of interest or equivalence margin was fixed; nonsignificance does not establish equivalence, and significance does not establish practical usefulness.

2 Scope and procedures

These analyses use the 25 primary targets and saved responses from the completed experiment. They are post hoc and descriptive. No new LLM requests were made. Original predictions, token totals, statistical tests, and reference tables are preserved. The alternative pipelines below are not substituted into the original eight conditions.

The matched direct-anchor variant uses the saved direct response at the exact configurations queried in the paired RNM request, preserving its row order and LLM-selected family. Pairing fixes node, replication, prompt, and evidence condition. Family selection by anchor fit searches WMEAN, WMIN, WMAX, and MIXMINMAX with the original optimizer and chooses the lowest mean anchor TV without reference access. Stronger search retains the original family and uses 16 initial starts plus the incumbent, 80 sweeps per step, and steps 2^{-2} through 2^{-14}. Seed 42, the 10^{-12} improvement threshold, five-point integration, and influence directions are retained. The incumbent remains a candidate. Expanded bounds change mean log ratios to $[-12, 12]$, minimum/maximum weights to $[1, 100]$, and variance to $[10^{-6}, 4]$, retaining the stronger-search candidate.

Reference-informed diagnostics separately fit complete reference tables. Each node considers all four families with stronger search and frozen signs/bounds, then expanded bounds, then all 2^k parent-direction combinations with expanded bounds. Selection minimizes complete-reference TV; these are in-sample best-found approximations, not predictions on hidden configurations or certified global optima. The no-flip expanded candidate is retained in the direction search. Family order breaks exact ties. Numerical plateaus can contain different parameters, so low loss does not establish parameter recovery.

3 Alternative pipelines fitted to LLM judgments

The table uses the original primary non-anchor rows and macro-aggregation: equal targets within BNs and equal BN versions within the Chrome family; prompt/evidence cells and five replications receive equal weight. The 14 Scrum and 11 Chrome targets remain separate. Uniform probabilities supply a reference-blind uninformed baseline. No additional p-values are calculated.

Pipeline	Scrum TV	Chrome TV
Original direct	0.3184	0.2223
Original RNM	0.2761	0.2517
Uniform	0.4525	0.3739
Direct anchors, paired RNM family	0.2924	0.2456
RNM anchors, family by anchor fit	0.2787	0.2413
RNM anchors, stronger search	0.2761	0.2517
RNM anchors, expanded bounds	0.2761	0.2530

4 Interpretation of alternative fits

Could the result mainly reflect different LLM answers or an insufficient parameter search? Fitting direct judgments on the paired RNM anchors still improves Scrum relative to the original direct pipeline (0.3184 to 0.2924) and worsens Chrome (0.2223 to 0.2456). Selecting the aggregation family by its fit to LLM anchors narrows Chrome's disadvantage to 0.2413, but does not reverse it. Stronger searches and wider parameter bounds make little difference. These post-hoc checks support a mismatch in the tested reconstruction pipeline rather than a simple shortage of optimization effort; they do not isolate every component.

Influence assumptions also matter. Four Chrome targets exhibit mean-direction reversals and six exhibit stochastic-order reversals under the specified parent alignment. Scrum has two targets with reversals. Even fitting directly to complete reference tables yields a higher best-found error for Chrome than Scrum (0.1592 versus 0.0560); allowing wider bounds and reversed directions lowers these to 0.1322 and 0.0260. This reference-informed diagnostic supports a compatibility explanation, without establishing a global optimum or a lower bound on RNM error.

5 Reference-informed approximation diagnostics

The following table reports complete-table TV, including fitted configurations. These values are not comparable to held-out predictive validation or to a mathematical error floor.

Reference-informed search	Scrum TV	Chrome TV
All families, stronger search	0.0560	0.1592
Expanded bounds	0.0559	0.1589
Expanded bounds and direction enumeration	0.0260	0.1322

6 Secondary metrics and replication differences

Metric	Scrum D	Scrum RNM	Chrome D	Chrome RNM
TV	0.3184	0.2761	0.2223	0.2517
Jensen–Shannon (bits)	0.1331	0.1066	0.0905	0.1104
Ordinal CDF	0.1887	0.1588	0.1809	0.2000

The five original replication differences (RNM minus direct TV) are:

Replication	Scrum	Chrome
1	-0.04538	+0.03410
2	-0.03414	+0.02773
3	-0.04732	+0.02682
4	-0.04155	+0.02935
5	-0.04314	+0.02907

7 Illustrative conditional distributions

For each stratum, we select the target nearest the unweighted median per-target RNM-minus-direct non-anchor TV effect, breaking ties by identifier. We show its all-middle aligned anchor and average the 20 saved distributions per method. This transparent rule does not select the largest benefit or harm. Mean probability vectors are illustrative; their TV need not equal mean trial TV.

7.1 All contributors

Number of distinct component contributors.

Parents: Component changes = [11, 51); Distinct reviewers = [3, 7).

Child state order: [1, 3), [3, 5), [5,102]. The target's mean non-anchor effect is +0.0193. All 20 RNM trials select WMEAN.

Distribution	State 1	State 2	State 3
Reference	0.356	0.421	0.223
Direct	0.261	0.358	0.382
Raw RNM	0.213	0.396	0.391
Fitted RNM	0.231	0.537	0.231

At this row, mean trial TV is direct: 0.1587, raw rnm: 0.1691, fitted rnm: 0.1318. These anchor illustrations do not establish a universal pattern for non-anchor rows.

7.2 Team Orientation

Mutual trust, respect and motivation to prioritize team goals over isolated individual goals.

Parents: Personal Attributes = Medium; Expertise = Medium.

Child state order: Low, Medium, High. The target's mean non-anchor effect is −0.0306. All 20 RNM trials select WMEAN.

Distribution	State 1	State 2	State 3
Reference	0.053	0.927	0.020
Direct	0.237	0.501	0.263
Raw RNM	0.206	0.585	0.210
Fitted RNM	0.207	0.586	0.207

At this row, mean trial TV is direct: 0.4260, raw rnm: 0.3425, fitted rnm: 0.3414. These anchor illustrations do not establish a universal pattern for non-anchor rows.

8 Qualitative evidence packets

All 61 retained nodes have nonempty packets containing 1–3 cards. Card text totals 24–103 whitespace-separated words per node, with median 72. The same node packet is reused across methods, prompt variants, and replications. The following are the complete cards for the two illustrative targets. Both evidence arms also receive the child and parent definitions.

8.1 All contributors

Component history. Changes are aggregated at component-directory level. Size, change volume and contributor counts describe different aspects of development activity; one does not uniquely determine another.

Contributor roles. Contributors are classified by their share of component changes. Minor and major contributors partition the contributor set. Holding total contributors fixed, a larger minor-contributor count leaves fewer major contributors.

Review measures. Review activity can include discussion, approval and revisions of a patch. A patch revision during review is not itself a confirmed post-release defect. Author-only approval and review without discussion describe distinct practices.

8.2 Team Orientation

Mutual support. Team orientation involves mutual respect and giving team goals priority. Constructive relationships and confidence in colleagues help members work together toward those goals.

Knowledge coverage. Expertise includes collective knowledge and overlapping capabilities. Being able to perform parts of another member's work supports flexibility when circumstances demand coverage.

Assessment scope. The assessment is intended to support retrospective discussion. Project-specific practices and metrics can be added to a general model. A contextual ordinal rating does not have a universal numerical threshold across organizations.

9 Target-level primary differences

9.1 Scrum

Network	Target	Direct	RNM	Δ
Scrum	Teamwork Quality	0.3050	0.3095	+0.0044
Scrum	Increment quality	0.3306	0.3510	+0.0204
Scrum	Customer satisfaction	0.1131	0.0694	-0.0437
Scrum	Cohesion	0.2471	0.1018	-0.1452
Scrum	Self-Organizing	0.3491	0.3487	-0.0004
Scrum	Collaboration	0.3671	0.2777	-0.0894
Scrum	Team Orientation	0.3869	0.3563	-0.0306
Scrum	Communication	0.4366	0.3707	-0.0660
Scrum	Coordination	0.1500	0.1284	-0.0216
Scrum	Continuous backlog refinement	0.2454	0.1708	-0.0746
Scrum	Sprint execution quality	0.3522	0.3316	-0.0206
Scrum	Requirements volatility	0.5307	0.4614	-0.0692
Scrum	Product backlog quality	0.3366	0.3062	-0.0304
Scrum	Sprint planning quality	0.3074	0.2821	-0.0253

9.2 Chrome

Network	Target	Direct	RNM	Δ
chrome_40	Distinct reviewers	0.1189	0.1419	+0.0229
chrome_40	Major contributors	0.0676	0.0728	+0.0052
chrome_40	Minor contributors	0.3389	0.3860	+0.0471
chrome_40	Prior defects	0.1728	0.1240	-0.0487
chrome_40	Patch revisions during review	0.3301	0.3146	-0.0156
chrome_40	Self-approved changes	0.2133	0.2219	+0.0086
chrome_42	All contributors	0.2139	0.2332	+0.0193
chrome_42	Major-author ownership	0.4344	0.4255	-0.0089
chrome_42	Major contributors	0.0000	0.1565	+0.1565
chrome_42	Minor contributors	0.3696	0.4561	+0.0866
chrome_42	Self-approved changes	0.1706	0.1949	+0.0243

These target means describe this sample; targets and rows are not independent experimental replications.

10 Version differences and leave-one-node-out diagnostics

These post-hoc descriptive checks reuse the original primary non-anchor scores, without refitting or new hypothesis tests. Each omission removes one primary target and re-averages the remaining targets within its BN; the two Chrome versions retain equal weight. Prompt/evidence cells and the five replications retain equal weight. Predictions, references, non-anchor row sets, and the original primary sample remain unchanged.

The version-level means are:

Network	Targets	Direct TV	RNM TV	Δ
Chrome 40	6	0.2069	0.2102	+0.0033
Chrome 42	5	0.2377	0.2933	+0.0556
Scrum	14	0.3184	0.2761	-0.0423

Chrome's aggregate disadvantage is concentrated in Chrome 42. No individual omission changes the original direction in any of the five replication blocks. The ranges below are descriptive minima and maxima across omissions, not confidence intervals.

Stratum	Omissions	Minimum Δ	Maximum Δ
Scrum	14	-0.0471	-0.0344
Chrome	11	+0.0168	+0.0375

Major Contributors in Chrome 42 has per-target difference +0.1565 and original macro-weight 0.1. Its weighted contribution is +0.01565, or 53.2% of the net Chrome contrast; this is an arithmetic contribution, not explained variance. Omitting it and reweighting within Chrome 42 leaves Δ = +0.0168, positive in every block. The node remains in all original analyses.

10.1 Scrum single-node omissions

Omitted target	Network	Remaining macro Δ
Teamwork Quality	Scrum	-0.0459
Increment quality	Scrum	-0.0471
Customer satisfaction	Scrum	-0.0422
Cohesion	Scrum	-0.0344
Self-Organizing	Scrum	-0.0455
Collaboration	Scrum	-0.0387
Team Orientation	Scrum	-0.0432
Communication	Scrum	-0.0405
Coordination	Scrum	-0.0439
Continuous backlog refinement	Scrum	-0.0398
Sprint execution quality	Scrum	-0.0440
Requirements volatility	Scrum	-0.0402
Product backlog quality	Scrum	-0.0432
Sprint planning quality	Scrum	-0.0436

10.2 Chrome single-node omissions

Omitted target	Network	Remaining macro Δ
Distinct reviewers	Chrome 40	+0.0274
Major contributors	Chrome 40	+0.0292
Minor contributors	Chrome 40	+0.0250
Prior defects	Chrome 40	+0.0346
Patch revisions during review	Chrome 40	+0.0313
Self-approved changes	Chrome 40	+0.0289
All contributors	Chrome 42	+0.0339
Major-author ownership	Chrome 42	+0.0375
Major contributors	Chrome 42	+0.0168
Minor contributors	Chrome 42	+0.0255
Self-approved changes	Chrome 42	+0.0333

11 Complete-table family means and anchor fitting contrasts

Complete-table TV averages are Scrum 0.2605 direct versus 0.2334 RNM; Chrome 0.2029 versus 0.2278; APS 0.0964 versus 0.1155; SESSI 0.1160 versus 0.1143; and software logs 0.2927 versus 0.4034. The equal-family external SE average is 0.1770 versus 0.2153. Excluding SESSI and reweighting gives 0.1973 versus 0.2489. These endpoints include all 61 targets and are distinct from the primary non-anchor comparison.

On identical anchor rows of primary targets, fitting changes Scrum TV by −0.0028 (pointwise 95% CI [−0.0053, −0.0003], raw and Holm t $p = 0.0352$), and Chrome TV by +0.0503 (CI [0.0430, 0.0576], raw $p = 4.46 \times 10^{-5}$, Holm $p = 8.93 \times 10^{-5}$). Both adjusted exact sign-test values are 0.125. These comparisons use five replication blocks; they do not decompose non-anchor error additively.

12 Complete condition means and statistical results

These tables provide the exact condition means, paired contrasts, and computational totals summarized in the main article. Primary comparisons use non-anchor rows; support comparisons use complete tables. The analysis unit is the five replication blocks.

Table 1. Primary non-anchor TV, averaged across prompt/evidence conditions and five replications. Δ is RNM minus direct. Intervals are pointwise; Holm adjustment covers the two strata. Exact sign tests give raw $p = 0.0625$ and adjusted $p = 0.125$ for each.

Reference	Targets	Direct	RNM	Δ	95% CI for Δ	Raw t p	Holm t p
Scrum	14	0.3184	0.2761	−0.0423	[−0.0486, −0.0360]	4.83×10^{-5}	4.83×10^{-5}
Chrome	11	0.2223	0.2517	+0.0294	[+0.0259, +0.0329]	1.99×10^{-5}	3.97×10^{-5}

Table 2. Mean TV for all eight conditions. Primary columns use non-anchor rows of the 14 Scrum and 11 Chrome targets. Complete-table columns use all rows of 25 Scrum targets and 36 external SE targets; the latter averages four publication families, not only Chrome. No cross-stratum accuracy mean is formed.

Method	Prompt	Evidence	Scrum primary	Chrome primary	Scrum complete	External SE complete
Direct	Baseline	None	0.3180	0.2214	0.2620	0.1773
Direct	Baseline	RAG	0.3226	0.2194	0.2660	0.1813
Direct	Structured	None	0.3150	0.2259	0.2548	0.1751
Direct	Structured	RAG	0.3181	0.2225	0.2594	0.1743
RNM	Baseline	None	0.2792	0.2486	0.2353	0.2145
RNM	Baseline	RAG	0.2745	0.2539	0.2346	0.2166
RNM	Structured	None	0.2729	0.2509	0.2302	0.2152
RNM	Structured	RAG	0.2778	0.2534	0.2335	0.2147

Table 3. Diagnostics on identical anchor configurations of primary targets. Direct, raw RNM, and fitted RNM errors are against the reference. Fit residual compares fitted and raw RNM judgments. All use primary aggregation weights.

Quantity	Scrum	Chrome
Direct anchor TV	0.2914	0.2507
Raw RNM anchor TV	0.2790	0.2491
Fitted anchor TV	0.2762	0.2994
Fit residual TV	0.0678	0.1253
Reconstructed non-anchor TV	0.2761	0.2517

Table 4. RQ2: complete-table TV differences, averaging the other prompt/evidence factor equally. "Prompt" means structured minus baseline; "RAG" means RAG minus no RAG. External SE comprises four publication families. Negative differences favor the added support. Holm adjustment covers eight contrasts.

Reference stratum	Method	Contrast	Δ	Pointwise 95% CI	Raw t p	Holm t p
Scrum	Direct	Prompt	−0.00692	[−0.00843, −0.00541]	0.000220	0.001761
Scrum	Direct	RAG	+0.00430	[+0.00289, +0.00572]	0.001082	0.007575
Scrum	RNM	Prompt	−0.00313	[−0.00943, +0.00318]	0.240329	1.000000
Scrum	RNM	RAG	+0.00132	[−0.00237, +0.00500]	0.377615	1.000000
External SE	Direct	Prompt	−0.00462	[−0.01205, +0.00282]	0.159742	0.958452
External SE	Direct	RAG	+0.00163	[−0.00412, +0.00738]	0.475067	1.000000
External SE	RNM	Prompt	−0.00060	[−0.00798, +0.00678]	0.832498	1.000000
External SE	RNM	RAG	+0.00085	[−0.00556, +0.00726]	0.731511	1.000000

Table 5. RQ3: observed tokens including repairs. Totals span five replications; the difference and its interval refer to *one replication's* paired method totals. Workloads combine strata for accounting only, not for accuracy inference. Holm adjustment covers four contrasts. All exact sign tests give raw $p = 0.0625$, adjusted $p = 0.25$.

Targets	Counter	Direct	RNM	Saved	Mean block Δ [95% CI]	Raw t p	Holm t p
All 61	Output	341,153	299,798	12.12%	−8,271 [−9,417, −7,125]	0.000037	0.000110
All 61	Input + output	14,697,090	14,607,517	0.61%	−17,915 [−39,331, +3,502]	0.080921	0.080921
Primary 25	Output	213,972	153,463	28.28%	−12,102 [−13,111, −11,093]	0.000005	0.000019
Primary 25	Input + output	6,384,519	6,196,992	2.94%	−37,505 [−58,833, −16,178]	0.008146	0.016292